MIL-P-43244A
14 March 1969
SUPERSEDING
MIL-P-43244
30 June 1964

MILITARY SPECIFICATION

PUNCHING MACHINE, PAPER, ELECTRIC MOTOR DRIVEN

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 This specification covers one type and size electric motor driven, paper-punching machine having a sheet width capacity of not less than 24 inches and a pile height capacity of not less than 3/8 inch (see 6.1 and 6.4).

* 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| CC-M-636 | - | Motor, Alternating Current, (Fractional Horsepower). |
| MMM-A-260 | - | Adhesive, Water-Resistant, (For Sealing Waterproofed Paper). |
| PPP-B-621 | - | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-1055 | - | Barrier Material, Waterproofed, Flexible. |

MILITARY

| | | |
|---|---|---|
| MIL-V-173 | - | Varnish, Moisture-and-Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment). |
| MIL-P-514 | - | Plates, Identification, Instruction and Marking, Blank. |
| MIL-P-3684 | - | Printing, Duplicating, and Book Binding Equipment, Packaging Of. |

FSC 3610

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement Of.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H-28 - Screw Threads for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402).

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Mass. 02110).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

* 3. REQUIREMENTS

3.1 First article.- When specified (see 6.2), before production is commenced, a preproduction sample of the finished punching machine, conforming to this specification, shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. Approval of the preproduction sample authorizes commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for manufacture of the production items.



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3.2 Standard product.- The punching machine covered by this specification shall be the regular commercial product of the manufacturer, except for changes necessary to comply with specification requirements. All machines furnished under any one contract or purchase order shall be identical throughout, including parts and assemblies thereof.

3.3 Standards compliance.-

3.3.1 Safety standard.- The end item shall conform to the applicable requirements of NFPA - 70.

3.3.2 Certification.- Prior to approval of the preproduction sample or, if none is submitted, prior to approval of the first shipment, the supplier shall submit to the contracting officer a certified statement that the end item conforms to the applicable requirements of NFPA - 70.

3.4 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer for paper punching machines, provided the completed items comply with all provisions of this specification.

3.5 Design and construction.- The machine shall be a floor mounted, electric motor driven, paper punching machine having a wood or plastic surface work table. The machine shall be designed to accommodate removable punching heads for round, open, and various other shape hole punches and dies. Power shall be transmitted from the electric motor to the punching heads by means of a foot treadle activated, non-repeating, power transmission device. The machine shall be furnished fully equipped, and shall include the punching heads with punches and dies specified herein; an adjustable side gauge, graduated in 1/16-inch increments; a removable chip receptacle; work light; and all features regularly furnished commercially with the manufacturer's basic commercial machine.

3.5.1 Capacities and dimensions.- Capacities and dimensions shall be as indicated in table I.

TABLE I.- Capacities and dimensions (inches, except as shown).

| | |
|---|---|
| Floor area required (maximum) | 41 by 33 |
| Capacity: | |
| Hole punches, round (minimum range) | 13/32 to 1/2 |
| Hole spacing (center to center minimum range) | 0 to 20 |
| Margin adjustment | 0 to 1-7/8 |
| Punching thickness | 0 to 3/8 |



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TABLE I.- Capacities and dimensions (inches, except as shown). (cont'd)

| | |
|---|---|
| Work table (± 2 inches): | |
| Depth | 14 |
| Length | 34 |
| Height (from base) | 38 |
| Sheet size capacity (± 2 inches) | 26 |
| Weight (net, lbs. max.) | 800 |

3.5.2 Electrical.- Unless otherwise specified (see 6.2), the machine shall be designed for operation on a 208 volt, 60 hertz, 3 phase power source. The electric motor shall conform to CC-M-636. The complete wiring circuit shall originate from marked terminals in an outlet box, and shall be grounded. Wiring shall be neat, sturdy, and as short as practicable, except that sufficient slack shall be provided to prevent stress on wires and connections. Wires shall be protected against abrasion where necessary and shall not be bent sharply around corners or edges that might cut or abrade the insulation. Where wires run through holes in metal, the insulation shall be protected with grommets or bushings. All conductors shall be continuous between outlets and there shall be no splice or tap within a raceway. Electrical connections shall be mechanically secured.

3.5.2.1 Fungus and moisture resistance.- When specified (see 6.2), the electrical circuitry, including all components and connections, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 with the following exceptions:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated. When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 0.001 inch to component or element surfaces previously cleaned and prepared.

3.5.2.2 Electromagnetic interference control.- When specified (see 6.2), the punching machines shall conform to the electromagnetic characteristics requirements and test limits for class IIB equipment as specified in MIL-STD-461 (see 4.4.1).

3.5.3 Screw threads.- All screw threads shall conform to Handbook H-28.



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3.5.4 Lubrication.- All parts requiring lubrication shall be provided with appropriate lubrication fittings, located in a protected, readily accessible location.

3.5.5 Maintainability.- All parts subject to periodic maintenance, repair, or adjustment shall be accessible for service without the need for complete disassembly of adjacent parts or assemblies.

3.6 Performance characteristics.- The machine shall punch one set of holes, without repeating, for each depression of the foot treadle. At the conclusion of each punching stroke the punches shall return to the "start" position. The machine shall consistently punch accurately placed, clean, sharp, distortion-free holes in paper, when tested in accordance with 4.4.2.

3.7 Punch heads, punches, and dies.- Punch heads, in the quantity specified, and with punches and dies in the size and shape specified, shall be furnished with each machine (see 6.2).

3.8 Finish.- The machines shall be given the same finish as normally applied by the manufacturer on his commercial machines.

3.9 Marking for identification.- The machine shall be marked for identification in accordance with MIL-STD-130 on an identification plate conforming to MIL-P-514. The plate shall be securely attached to the machine.

3.10 Spare parts, tools, and technical publications.- Spare parts, tools, and technical publications shall be as specified (see 6.2).

3.11 Workmanship.- The end item shall be free of dents and fractures, and shall show no signs of warping or misalignment. All wires shall be unbroken, unfrayed, and completely secured to their terminals. All welding, when applicable, shall be complete, uniform, and free from burn holes, scales, blisters, cracks, slag inclusion, or pock marks. All flux deposits shall be removed.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects in table II, dimensions specified, and tested as specified in 4.4.1 and 4.4.2, as applicable.



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4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 Process examination.- When applicable, examination shall be made during an appropriate phase of the manufacturing process to determine compliance with the fungus and moisture proofing requirements in 3.5.2.1. Whenever nonconformance is noted, correction shall be made to affected items and process.

4.3.3 End item inspection.- The lot shall be all punching machines offered for inspection at one time. The sample unit shall be one punching machine.

4.3.3.1 End item visual examination.- The end item shall be examined for defects indicated in table II. The inspection level shall be II. The acceptable quality level (AQL) shall be 4.0, expressed in terms of defects per hundred units.

TABLE II.- Visual defects, end item.

| Examine | Defect |
|---|---|
| Materials | Not as regularly used by the manufacturer. |
| Design and construction | Not designed for floor mounting.<br>Not electric motor driven.<br>Work table not wood or plastic surfaced.<br>Punching heads not removable.<br>Any component or assembly missing or incomplete.<br>Chip receptacle not removable. |
| Electrical | Insufficient or excessive slack in wiring.<br>Does not originate from marked terminals in an outlet box.<br>Not grounded.<br>Not protected against abrasion.<br>Conductors not continuous within raceway (when applicable).<br>Connections not mechanically secured. |



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TABLE II.- Visual defects, end item (cont'd)

| Examine | Defect |
|---|---|
| Workmanship | Screw threads not in accordance with Handbook H-28.<br>Lubrication fitting not accessible.<br>Any component loose, inoperable, binds, or fails to perform required function.<br>Parts subject to adjustment, maintenance, or replacement, not accessible.<br>Any component or assembly bent, warped, cracked, or otherwise damaged.<br>Any burrs or sharp edges.<br>Weld burned through, excess splatter not removed, blister, pock mark.<br>Weld incomplete, cracked, slag inclusion, or porous.<br>Flux not removed.<br>Fastening device broken, stripped, loose. |
| Technical publications | Missing, incomplete, incorrect, or illegible, when applicable. |
| Spare parts and tools | Any part or tool missing, damaged, or not as specified, when applicable. |
| Marking for identification | Missing, incorrect, illegible, not on prescribed plate, or not properly attached. |
| Finish | Not manufacturer's regular, commercial finish.<br>Rust, scale, flaking, or foreign matter imbedded. |

4.3.3.2 Dimensional examination.- The end item shall be examined for compliance with dimensions, capacities, and weight specified (see 3.5 and table I). Any evidence of noncompliance shall be classified a defect. The inspection level shall be S-2. The AQL shall be 4.0, expressed in terms of defects per hundred units.



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4.3.3.3 Testing of the end item.- When a preproduction sample is not required, the initial production unit shall be tested in accordance with 4.4.1, when applicable. In addition, the end item shall be tested as specified in 4.4.2, using inspection level S-2 and an AQL of 4.0, expressed in terms of defects per hundred units.

4.3.3.4 Code and standards compliance.- Proof of compliance with NFPA - 70 (see 3.3.1 and 3.3.2) shall be made available to the Government representative.

4.3.4 Examination of preparation for delivery.- Examination of preservation, packaging, and packing shall be conducted in accordance with MIL-P-3684, with additional examination added, as required, to determine compliance with the exceptions to MIL-P-3684, as specified in section 5 of this specification.

4.4 Tests.-

4.4.1 Electromagnetic interference control.- When electromagnetic interference control is required, the preproduction sample or initial unit of production shall be tested by the supplier in accordance with Test Methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or his private testing organization. The supplier shall furnish the contracting officer for approval, the Interference Control Plan, EMI/EMC Test Plan, and the Electromagnetic Compatibility Test Report required in MIL-STD-461. Upon approval of the test report by the contracting officer, and provided all other requirements of the specification are met, the preproduction sample, or initial unit of production, as applicable, shall be used as a model for all production units.

4.4.2 Performance.- The punching machine shall be tested in two phases as follows:

Phase I. The machine shall be equipped with 4 round hole or open hole punch heads having 13/32-inch diameter punches with matching dies. One sheet of 9 pound manifold paper, 8 inches wide by at least 22 inches long, shall be placed under the punch heads. The side gauge shall be set to allow punching of holes at a maximum distance between centers. The paper shall be placed so that the holes shall be punched the minimum possible distance from the margin. The motor shall then be energized and the treadle operated. The punched holes shall be examined to determine compliance with 3.6. The machine shall then be tested for performance with a 3/8 inch lift of 8 by 22 inches minimum, twenty-pound bond paper. This test shall be conducted in the same manner as for the single sheet except that the paper shall be placed



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so that the holes shall be punched the maximum possible distance from the margin. The lift of paper shall then be punched and the holes examined to determine compliance with 3.6.

Phase II. Repeat Phase I except use 4 round hole or open hole punch heads having 1/2-inch diameter punches with matching dies.

* 5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for level A, B or C shall be in accordance with MIL-P-3684 (see 6.2), except that detached components, punches and dies not installed on the machine, and tools may be packaged in a paperboard box in lieu of fiberboard box, and for level A pack, the contents of each nailed wood shipping container conforming to PPP-B-621 shall be covered with a full-height shroud made of waterproof barrier material conforming to class E-2, H-1, H-5, or M-1 of PPP-B-1055. All seams of the shroud shall be sealed with a 3/4-inch minimum width continuous seam of water resistant adhesive conforming to MMM-A-260.

* 6. NOTES

6.1 Intended use. The punching machine covered by this specification is intended to be used in print shops and book binderies for the purpose of accurately punching round and other shaped holes in the margins of paper for subsequent retention in the form of booklets or pamphlets, in ring binders and other similar binding media. When equipped with special punch heads, the machine may be used to cut and perforate paper.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) First article.- When a preproduction sample is required (see 3.1).
(c) Electrical power source characteristics if other than specified (see 3.5.2).
(d) When fungus and moisture resistance protection is required (see 3.5.2.1).
(e) When electromagnetic interference control is required (see 3.5.2.2).
(f) Quantity of punch heads required (see 3.7).
(g) Quantity, size and shape punches and dies required (see 3.7).
(h) Spare parts, tools and technical publications required (see 3.10).
(i) Selection of applicable levels of preservation, packaging and packing (see 5.1).



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6.3 Electromagnetic compatibility.- When applicable, the contracting officer should submit a copy of the Interference Control Plan, the EMI/EMC Test Plan and the test report required (see 4.4.1) to the Commanding General, U. S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, N. J. 07703 for approval.

6.4 Classification change.- The type I, foot driven machine, covered in MIL-P-43244 dated 30 June 1964, has been deleted in this revision. The type II machine formerly covered by MIL-P-43244 is covered herein.

6.5 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 84

Review activities:

Army - EL
Air Force - 84, 85

Preparing activity:

Army - GL

Project No. 3610-0153



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Punching Machine, Paper, Electric Motor Driven MIL-P-43244A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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